

★ ALTERNATING + PAIRS/GROUPS ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 1 of 12

► Meaning in one line

Zigzag , blocks , sums.

► Key points

Point	Example
split.	M, 1, O, 3, Q, 5 → S, 7.
blocks + sums.	AB, DE, GH → JK (+3 block).

★ IMPORTANT FOR EXAM

Coding (shifts)

⚠ TRAP - read twice

Single-track + 'random' + block-miss

✓ **Try yourself:** Next: B1, D4, G9, K16, ? **Answer:** Letters +3, squares: P25.

★ SHIFT + POSITION CODING ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 2 of 12

► Meaning in one line

Move , mirror , sums.

► Key points

Point	Example
+n/-n + wrap.	CODE (+2) → EQFG.
A1-Z26/reverse + sums.	BAD = 2+1+4 = 7; reverse-Z = 1? Z→A=1...

★ IMPORTANT FOR EXAM

Substitution + word-number

⚠ TRAP - read twice

A0 + 'Z+1' + reverse-mix + sums-miss

✓ **Try yourself:** CAT (-1)? Sum: DOG? **Answer:** BZS; 4+15+7 = 26.

★ SUBSTITUTION + WORD-NUMBER CODING ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 3 of 12

► Meaning in one line

Replace , map , consistent.

► Key points

Point	Example
map + apply-all.	if ATE = BXJ (+1 each) → EAT = FBV.
values + sums.	PEN = 16+5+14 = 35.

★ IMPORTANT FOR EXAM

Symbol + conditional + comparison

⚠ TRAP - read twice

One-letter + 'pattern obvious' + map-break + sums-miss

✓ **Try yourself:** If DOG = 26 (sums), CAT? **Answer:** 3+1+20 = 24.

★ SYMBOL + CONDITIONAL CODING ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 4 of 12

► Meaning in one line

Signs , rules , apply.

► Key points

Point	Example
sign-map + apply.	$\$ = A, \# = T \rightarrow \$\# = AT.$
conditions in order.	vowels +2, consonants -1.

★ IMPORTANT FOR EXAM

Comparison-trap + relations

⚠ TRAP - read twice

One-sign + 'code cracked' + condition-order-miss

✓ **Try yourself:** Rule + apply + condition-order. Steps? **Answer:** Map-all + apply.

★ CODING COMPARISON TRAP ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 5 of 12

► Meaning in one line

Two-codes , common , difference.

► Key points

Point	Example
Compare + common + difference + isolate.	-
Ex: CAT = 24, BAT = 22 → C-B = 2 (3-2 = 1? use sums: C=3,B=2 diff 1...). Solve via full sums.	-

★ IMPORTANT FOR EXAM

Family relations

⚠ TRAP - read twice

Single-code + 'solved' + common-miss

✓ **Try yourself:** Method + isolate. Steps? **Answer:** Compare + common + difference.

★ FAMILY TREE + DIRECT RELATIONS ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 6 of 12

► Meaning in one line

Male/female , generations , lines.

► Key points

Point	Example
symbols + generations.	draw + count.
terms + side.	maternal-uncle = mother's brother.

★ IMPORTANT FOR EXAM

Generation + coded + pointing

⚠ TRAP - read twice

No-tree + 'relation obvious' + side-mix + generation-miss

✓ **Try yourself:** Mother's sister's husband? **Answer:** Maternal-uncle (aunt's husband).

★ GENERATION + CODED RELATIONS ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 7 of 12

► Meaning in one line

Count , gaps , decode , substitute.

► Key points

Point	Example
count gaps.	grandfather $\rightarrow +2$; grandson $\rightarrow -2$.
decode + substitute.	$A+B$ = siblings; $A \times B$ = parent $\rightarrow$ solve.

★ IMPORTANT FOR EXAM

Pointing + ONLY + gender-trap

⚠ TRAP - read twice

Gap-miss + 'generation obvious' + decode-half

✓ **Try yourself:** $A+B$ siblings, $B+C$ siblings, $A+C$? **Answer:** All siblings (transitive).

★ POINTING + ONLY/GENDER TRAPS ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 8 of 12

► Meaning in one line

Words , logic.

► Key points

Point	Example
words + logic.	'only son of my father' = self/brother.
ONLY + unknown-gender.	'his mother's only daughter' = his sister (or self if female).

★ IMPORTANT FOR EXAM

Direction (compass + turns)

⚠ TRAP - read twice

Assume-gender + 'ONLY obvious' + words-rush

✓ **Try yourself:** 'Only daughter of my mother' (speaker male)? **Answer:** His sister.

★ COMPASS + TURNS ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 9 of 12

► Meaning in one line

N/E/S/W , face , left/right , 90/180.

► Key points

Point	Example
N/E/S/W + face.	face East, left = North.
left/right + degrees + track.	North, right-right-about = South? $R=N \rightarrow E$, $R=E \rightarrow S$, $U=S \rightarrow N$.

★ IMPORTANT FOR EXAM

Diagonal + path + shortest

⚠ TRAP - read twice

Face-miss + 'left = west' + turn-count + no-track

✓ **Try yourself:** Face South, left, then right. Final? **Answer:** East, then South.

★ DIAGONAL + PATH TRACING ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 10 of 12

► Meaning in one line

NE/SE/SW/NW , 45° , steps , map.

► Key points

Point	Example
45° + diagonals.	NE + right-45 = E? NE→E (right) yes.
trace + turns.	N2, E3, S2 → 3 East of start.

★ IMPORTANT FOR EXAM

Shortest + relative + shadow

⚠ TRAP - read twice

Cardinal-only + '45° = 90' + trace-miss

✓ **Try yourself:** N5, E12, S5. Displacement + direction? **Answer:** 12 East (Pythagoras legs 0? N–S cancel).

★ SHORTEST + RELATIVE + SHADOW ★

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► Meaning in one line

Pythagoras , legs.

► Key points

Point	Example
legs + Pythagoras.	6E + 8N → 10 NE.
morning-shadow-West.	face sun (East AM) + shadow behind = West.

★ IMPORTANT FOR EXAM

Ranking (both-ends)

⚠ TRAP - read twice

Path-length + 'shortest = walk' + shadow-flip

✓ **Try yourself:** 3E + 4N shortest? Morning, sun front, shadow? **Answer:** 5 NE; behind = West.

★ RANK: BOTH ENDS + OPPOSITE ★

Reasoning | Ch02 Coding + Relations + Direction | Topic 12 of 12

► Meaning in one line

Left, right-1 = total , mirror.

► Key points

Point	Example
total = L+R-1.	7th left + 12th right = 18 total.
mirror = total-rank+1.	total 20, 7th left → 14th right.

★ IMPORTANT FOR EXAM

Between + interchange

⚠ TRAP - read twice

-1-miss + 'total = sum' + mirror-mix

✓ **Try yourself:** Total 30, 9th from left. From right? **Answer:** $30-9+1 = 22$ nd.